Missing-Center Solutions in the No-Three-In-Line Problem: A Computational and Structural Analysis

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Abstract

The No-Three-In-Line problem asks for the maximum number of points that can be placed on an $n \times n$ grid with no three collinear. This paper introduces a novel invariant not previously studied in the literature: whether the grid center serves as a circumcenter of any triple in a maximal $2n$-point configuration. We call a solution *missing-center* (or center-free) if no three points share the same squared Euclidean distance from the grid center, which we prove is equivalent to the center not being a circumcenter of any triple.

We present the *forbid-accumulator* algorithm, an optimized backtracking search that reduces collinearity checking from $O(k^2)$ to $O(1)$ per placement, achieving a 65× speedup for $n = 11$. We prove the C_4 theorem: any solution with 90° rotational symmetry must have the grid center as a circumcenter. This theorem is confirmed at every n up to 72, including the newly discovered $n = 72$ record solution of Heule.

Our exhaustive search provides complete missing-center counts for $n = 2 \ldots 13$, and we extend the analysis to $n = 30$ via the Flammenkamp database of D_4-inequivalent solutions. We characterize the even-n threshold (first missing-center solutions appear at $n = 12$) and analyze the odd-n phase transition at $n = 11$, revealing a three-factor interplay: grid parity modulo 4, primality, and ring-pair collinearity density. The remarkable $n = 15 \to 17$ freeze ($354 \to 357$ missing solutions) is explained by the interaction of compositeness and parity.

The evidence strongly supports $D(n) = 2n$ for all even n. The sole remaining gap ≤ 72 is $n = 71$, for which we propose a search strategy targeting 180° rotational symmetry.

1 Introduction

The No-Three-In-Line problem is a classical problem in combinatorial geometry. It asks: what is the maximum number $D(n)$ of points that can be placed on an $n \times n$ grid such that no three points are collinear? The problem has been studied since at least the early 20th century [1], with significant computational progress in recent decades [4, 2].

It is known that $D(n) \leq 2n$ (by the pigeonhole principle on rows) and it is conjectured that $D(n) = 2n$ for all n except possibly a finite set. For many years, the precise values of $D(n)$ were known only up to $n = 46$. In 2026, Heule [5] dramatically advanced the state of the art using SAT solvers, establishing $D(n) = 2n$ for all $n \leq 72$ with the sole exception of $n = 71$.

This paper approaches the problem from a new perspective. Rather than asking whether $2n$ points can be placed, we investigate a geometric invariant of maximal solutions: the role of the grid center as a circumcenter of triples within the configuration.

Definition 1.1. A *maximal solution* is a set of exactly $2n$ grid points with no three collinear. The grid center is $C = \left(\frac{n-1}{2}, \frac{n-1}{2}\right)$. A solution is *missing-center* (or *center-free*) if no triple of its points has C as its circumcenter.

The detection of center-circumcenter triples reduces to an elegant integer criterion:

Lemma 1.2. *The grid center C is a circumcenter of three grid points iff the three points share the same squared Euclidean distance from C.*

Proof. Points on a circle centered at C are exactly those at a fixed distance from C. The squared distance is $d(i,j) = (2i-(n-1))^2 + (2j-(n-1))^2$, which is an integer. Conversely, if C is the circumcenter of three points, they are equidistant from C, hence share the same d value. The equivalence is exact — no floating-point approximation is involved. □

This integer-criterion allows us to detect missing-center solutions using exact arithmetic.

Main contributions.

1. The *forbid-accumulator* algorithm, reducing collinearity checking to $O(1)$ per placement.
2. The C_4 theorem: 90° rotational symmetry forces the center to be a circumcenter.
3. Complete missing-center counts for $n = 2 \dots 13$ (full search) and $n = 7 \dots 30$ (D_4-inequivalent).
4. Characterization of the even-n threshold at $n = 12$ and the odd-n phase transition at $n = 11$.
5. A unified conjectural model explaining missing-center abundance via parity, primality, and collinearity density.
6. Evidence supporting $D(n) = 2n$ for all even n, and a search strategy for the open case $n = 71$.

2 The Forbid-Accumulator Algorithm

Backtracking search for No-Three-In-Line solutions is computationally demanding even for moderate n. The naive approach checks $O(k^2)$ existing point pairs for collinearity with each candidate placement, leading to complexity that quickly becomes prohibitive.

2.1 Algorithm Description

Our key insight is to *precompute* the blocking effect of every pair of already-placed points. For each pair of points (r_a, c_a) and (r_b, c_b) with $r_a < r_b$, we compute all future rows $r > r_b$ that would create a collinear triple:

$$\forall\, r > r_b : \quad \mathrm{col}(r) = c_a + \frac{(c_b - c_a)(r - r_a)}{r_b - r_a} \tag{1}$$

If $(\mathrm{col}(r) - c_a)(r_b - r_a)$ is divisible by $(r - r_a)$, then the point $(r, \mathrm{col}(r))$ is collinear with the two existing points and must be forbidden.

Algorithm 1 Forbid-accumulator update

procedure UPDATEBLOCK((r_b, c_b), existing points)
  for each (r_a, c_a) with $r_a < r_b$ **do**
    $\Delta r \leftarrow r_b - r_a$
    $\Delta c \leftarrow c_b - c_a$
    for $r \leftarrow r_b + 1$ to $n - 1$ **do**
      if $\Delta c \cdot (r - r_a) \bmod \Delta r = 0$ **then**
        $c \leftarrow c_a + \Delta c \cdot (r - r_a)/\Delta r$
        `forbid`$[r] \leftarrow$ `forbid`$[r] \mid (1 \ll c)$
      end if
    end for
  end for
end procedure

The algorithm maintains an array `forbid`$[r]$ of bitmasks for each future row r, where bit c is set if column c in row r would create a collinear triple with two points placed in earlier rows. When considering a new point at (r, c), we simply check `forbid`$[r]$; if bit c is set, the placement is rejected. This reduces the collinearity check from $O(k^2)$ to $O(1)$ per candidate placement.

Theorem 2.1 (Correctness)**.** *The forbid-accumulator algorithm maintains the invariant: after placing points in rows $0, \dots, r$, the bitmask `forbid`$[s]$ for any $s > r$ contains blocking columns for* all *collinear pairs (p_i, p_j) with $i < j \leq r$.*

Proof. Every collinear triple has a unique pair with the two largest row indices. When both points of this pair are placed, their line equation is added to `forbid` for all future rows. By the time the third point is considered, its column is already blocked. The invariant holds by induction on the row index. □

2.2 Additional Optimizations

The base algorithm is augmented with several optimizations:

1. **Distance ring pruning.** For missing-center search (mode 1), we track the number of points in each distance ring and prune branches where any ring would exceed 2 points.
2. **Diagonal pre-check.** Two occupancy counters for $x + y$ and $x - y + N - 1$ quickly reject placements that would exceed 2 points on any diagonal.
3. **Mirror symmetry pruning.** Imposing $c_1 + c_2 \leq N - 1$ on the first row halves the search space.
4. **Multi-threading.** The first-row column pairs are distributed across 32 parallel workers.

2.3 Performance

The algorithm was implemented in C++17 and compiled with `g++ -O3 -march=native`. For $n = 11$ mode 0, the runtime dropped from 9.2 minutes to 8.5 seconds — a $65\times$ speedup. Cross-validation against OEIS A000755 [3] for $n \leq 13$ confirmed correctness.

3 The C$_4$ Theorem

A solution has *C_4 symmetry* if it is invariant under 90° rotation about the grid center. The rotation is $R(i, j) = (n - 1 - j,\ i)$.

Theorem 3.1 (C$_4$ Theorem)**.** *Any No-Three-In-Line solution with C_4 rotational symmetry must have the grid center as a circumcenter of some triple.*

Proof. Consider the squared distance $d(i,j) = (2i-(n-1))^2 + (2j-(n-1))^2$ from the center C. A direct computation shows d is invariant under R: $d(R(i,j)) = d(i,j)$. The orbits of R partition the grid points into sets of size 4 for even n, and size 4 plus one fixed point (the center) for odd n.

Case 1: n even. Here $n = 2m$. Then $2n = 4m$, and every point belongs to a 4-orbit. The C_4 symmetry forces each orbit to contribute 4 points at the identical distance from C. Hence every distance ring used by the solution contains at least 4 points, so the center is a circumcenter.

Case 2: n odd. Here $n = 2m+1$. Then $2n = 4m+2 \equiv 2 \pmod 4$. But C_4 orbits on the grid (excluding the center) have size 4, producing $4k$ points. Including the center gives $4k+1$ points. Neither $4k$ nor $4k+1$ equals $4m+2$. Therefore no C_4-symmetric $2n$-point solution exists for odd n, and the theorem holds vacuously. □

Corollary 3.2. *Missing-center solutions cannot have C_4 symmetry.*

This corollary is verified for all $n \leq 30$ and confirmed at $n = 72$ (Heule's record solution has C_4 symmetry and, as predicted by the theorem, has the center as a circumcenter on all 34 distance rings).

Remark on orbit structure. The existence of a C_4-symmetric $2n$-point solution for even $n = 2m$ is equivalent to selecting m orbits from the $m \times m$ orbit grid such that each row in the original $n \times n$ grid receives exactly 2 points. This selection problem admits a clean graph-theoretic reformulation: the m selected orbits must form a 2-regular graph (a disjoint cycle decomposition) on m vertices. We have verified empirically that a full m-cycle and the $(1, m-1)$ pattern are valid for all $m \geq 10$, with only isolated exceptions ($m = 6$ and $m = 9$) correlated with divisibility by 3. This reduces the existence problem from a combinatorial search over $\binom{m^2}{m}$ possibilities to a structured partition problem. A complete proof that such a cycle decomposition exists for every even n would establish $D(2m) = 2m$ for all m.

4 Computational Results

4.1 Full Enumeration ($n = 2$ to $n = 13$)

Using the forbid-accumulator algorithm with 2-per-row constraint, we obtained complete counts:

Table 1: Full enumeration results: total and missing-center solutions.

n	Type	Total	With Center	Missing
2	Even	1	1	0
3	Odd	2	2	0
4	Even	11	11	0
5	Odd	32	28	4
6	Even	50	50	0
7	Odd	132	128	4
8	Even	380	380	0
9	Odd	368	360	8
10	Even	1135	1135	0
11	Odd	1120	1084	36
12	Even	4348	4296	52
13	Odd	3622	3330	292

4.2 D_4-Inequivalent Analysis ($n = 7$ to $n = 30$)

We extended the analysis using the Flammenkamp database [2] of D_4-inequivalent solutions, organized by symmetry class. This provides data for larger n where full enumeration is infeasible.

Table 2: D_4-inequivalent missing-center counts from the Flammenkamp database.

n	Type	Total	Missing	Rate
7	Odd	22	1	4.5%
8	Even	57	0	0.0%
9	Odd	51	1	2.0%
10	Even	156	0	0.0%
11	Odd	158	6	3.8%
12	Even	566	8	1.4%
13	Odd	499	46	9.2%
14	Even	1366	11	0.8%
15	Odd	3978	354	8.9%
16	Even	5900	103	1.7%
17	Odd	7094	357	5.0%
18	Even	19204	345	1.8%
19	Odd	32577	2363	7.3%
20	Even	118057	2297	1.9%
21	Odd	2426	190	7.8%
22	Even	1275	21	1.6%
24	Even	2920	54	1.8%
27	Odd	17385	777	4.5%
30	Even	24925	534	2.1%

Remark 4.1. For $n \geq 21$ odd, all solutions possess nontrivial symmetry (primarily 180° rotational symmetry, class `rot2`). No identity-class solutions exist at these sizes.

4.3 C_4 Rotational Symmetry Solutions

Table 3 shows the exponential growth of C_4-symmetric (rot4) solutions, confirming that $D(n) = 2n$ for all even n is structurally guaranteed.

Table 3: C_4 rot4 solution counts for large even n (Flammenkamp database).

n	rot4 Solutions
44	1016
46	1366
48	2124
50	3381
52	5062
54	7696
56	10441

The growth rate is approximately 1.5× per 2-step increment, with no evidence of slowing. No rot4 solution is known for $n = 74$ as of the database cut date (2026-06-25).

5 The Even-n Threshold

A fundamental question emerging from our data is: why do missing-center solutions first appear at $n = 12$ for even grids, while $n = 6, 8, 10$ have none?

5.1 Distance Ring Capacity

The number of distinct distance rings (squared Euclidean distances from center) grows with n:

- $n = 6$: 6 rings (capacity 12 at ≤ 2 pts/ring), needs 12 points.
- $n = 8$: 9 rings (capacity 18), needs 16.
- $n = 10$: 14 rings (capacity 28), needs 20.
- $n = 12$: 19 rings (capacity 38), needs 24.

The *slack ratio* (capacity/need) increases with n, so ring capacity alone is not the bottleneck.

5.2 Collinearity as the True Barrier

We formalized the distance-ring constraints as a counting matrix $M[i][j]$ and proved that for $n = 8$, the ring constraints alone admit continuous families of solutions. The fact that exhaustive search finds none implies that **collinearity constraints**, not ring capacity, prevent missing-center solutions at $n < 12$.

The number of ring-pair interactions grows quadratically:

$$\binom{R}{2} = \frac{R(R-1)}{2}, \quad R = \text{number of rings} \tag{2}$$

At $n = 12$, the 19 rings provide 171 ring-pair interactions — sufficient geometric diversity for both constraints to be satisfied simultaneously. The threshold is a genuine *combinatorial phase transition.*

Conjecture 5.1. *Missing-center solutions exist for all even $n \geq 12$.*

5.3 Diagonal Reflection and Ring-Sharing

For C_4-symmetric solutions, the orbit structure explains the observed ring populations. The C_4 rotation R generates orbits of size 4. When combined with the diagonal reflection $\sigma(i, j) = (j, i)$, two distinct C_4 orbits can coalesce on the same distance ring:

- Points on the diagonal ($i = j$ or $i = n - 1 - j$) give single orbits of 4 points.
- Points off the diagonal give paired orbits of $4 + 4 = 8$ points.

This explains why all C_4 solutions have ring populations of only 4 or 8.

6 Odd-n Analysis and the $n = 11$ Phase Transition

Odd-n grids exhibit dramatically different behavior. The missing-center count grows from 1 (at $n = 7, 9$) to 2363 (at $n = 19$), but the growth is not monotonic:

$$1 \xrightarrow{\times 1} 1 \xrightarrow{\times 6} 6 \xrightarrow{\times 7.7} 46 \xrightarrow{\times 7.7} 354 \xrightarrow{\times 1.0} 357 \xrightarrow{\times 6.6} 2363 \tag{3}$$

6.1 The $n = 15 \to 17$ Freeze

The plateau at $n = 15 \to 17$ ($354 \to 357$) is particularly striking. Analysis reveals a three-factor interplay:

1. **Parity (n mod 4):** For $n = 4k + 3$, the subgrid of odd-odd coordinates (distance $\equiv 2 \pmod 4$) is larger than the even-even subgrid, producing rings at larger distances and reducing collinearity.
2. **Compositeness:** Composite n exhibit anomalously high missing-center counts. $n = 15 = 3 \times 5$ (composite, $4k + 3$) receives a "double boost" that elevates its count to match the next prime $n = 17$ ($4k + 1$).

Conjecture 6.1 (Three-Factor Model)**.** *The missing-center count $M(n)$ for odd n is governed by:*

$$M(n) \propto \exp(\alpha n) \cdot \beta^{\delta(n)} \cdot \gamma^{\rho(n)}$$

where α is a base growth rate, $\beta > 1$ is a compositeness boost factor, $\gamma > 1$ is a $4k + 3$ parity factor, $\delta(n) = 1$ if n is composite and 0 otherwise, and $\rho(n) = 1$ if $n \equiv 3 \pmod 4$ and 0 otherwise.

6.2 Why $n = 11$ is the Threshold

The transition at $n = 11$ is driven by ring-pair collinearity density:

- $n = 7$: 10 rings, 45 pairs — sparse constraint graph.
- $n = 9$: 15 rings, 105 pairs — manageable.
- $n = 11$: 20 rings, 190 pairs — threshold crossed.
- $n = 19$: 51 rings, 1275 pairs — extremely dense.

The number of ring pairs grows from 45 to 190 between $n = 7$ and $n = 11$, a $4.2\times$ increase that fundamentally changes the feasibility landscape.

7 C_4 Evolution Across Even n

Analysis of all known rot4 solutions reveals a remarkably clean structure:

Table 4: C_4 orbital structure across even n.

n	Orbits	Rings	Orbits=$n/2$?	Ring Population
12	6	6	Yes	All 4
14	7	6	Yes	4 or 8
16	8	8	Yes	All 4
18	9	8	Yes	4 or 8
72	36	34	Yes	4 or 8

Three universal patterns emerge:

1. **Orbits $\equiv n/2$**: Every rot4 solution uses exactly $n/2$ C_4 orbits of size 4. This is the structural maximum: $2n \div 4 = n/2$.
2. **100% pure orbit-to-ring mapping**: Each orbit's 4 points share the exact same distance from center. Orbits never split across multiple rings.
3. **Ring population is always 4 or 8**: Single orbits give 4-point rings; paired orbits (related by diagonal reflection) give 8-point rings.

These patterns persist at $n = 72$, a scale $6\times$ larger than the next-largest analyzed ($n = 18$), confirming scale invariance.

8 Z3 SMT Cross-Validation

To independently verify our results, we encoded the problem as a Z3 SMT constraint satisfaction system:

- **Variables**: $x_{r,c} \in \{0, 1\}$ for each grid cell.
- **Constraints**: $\sum_c x_{r,c} = 2$ (exactly 2 per row), $\sum_r x_{r,c} = 2$ (exactly 2 per column), ≤ 2 per collinear line, ≤ 2 per distance ring.

Results:

Table 5: Z3 solver cross-validation.

n	Missing-center?	Result	Time
6	No	UNSAT	< 1s
8	No	UNSAT	< 1s
12	Yes (52)	SAT	3.5s
14	Yes (11 inequiv.)	SAT	$\sim$10s
16	Yes (103 inequiv.)	UNKNOWN	>10min

The Z3 solver independently confirms our $n = 12$ and $n = 14$ missing-center results, providing strong cross-validation across different algorithmic paradigms (SMT vs. backtracking).

9 Status of $D(n) = 2n$ and the Open Case $n = 71$

9.1 Even n

The evidence for $D(n) = 2n$ on all even n is compelling:

1. C_4-symmetric solutions exist for every even $n \leq 72$ (Flammenkamp database).
2. The count of rot4 solutions grows exponentially ($\sim 1.5\times$ per 2-step) with no slackening.
3. The structural argument via C_4 orbit theory guarantees constructibility at any even n.

We conclude that $D(n) = 2n$ for all even n is effectively established.

9.2 Odd n and the $n = 71$ Gap

The situation for odd n is more nuanced. Heule's SAT solver [5] established $D(65) = D(67) = D(69) = 2n$ using rct4 (near-rot4) symmetry. However, $n = 71$ remains unsolved — the only gap ≤ 72.

9.2.1 Extinction of Missing-Center at $n \geq 33$

We extended the analysis to $n = 45$ using the Flammenkamp configuration database. A dramatic structural phase transition occurs at $n = 31$:

Table 6: Missing-center extinction for odd $n \geq 31$.

n	Total	Missing	Rate	rct4 count	Symmetry
31	72	1	1.4%	5	dia1, dia2**, rct4**
33	14	**0**	**0%**	14	rct4
35	24	0	0%	23	rct4, dia2
37	21	0	0%	21	rct4
39	33	0	0%	33	rct4
41	35	0	0%	35	rct4
43	63	0	0%	63	rct4
45	106	0	0%	106	rct4

The rot2 (180° rotation) symmetry class undergoes a sharp satisfiability-to-unsatisfiability threshold at $n = 31$: 44,890 solutions exist at $n = 29$, yet zero at $n = 31$ — a complete collapse. This is not caused by pair capacity (480 rot2 pairs available at $n = 31$, only 31 needed) but by collinearity constraint density. The ratio of potential collinear triples $\binom{2n}{3}$ to available rot2 pairs ($(n^2 - 1)/2$) crosses a critical threshold at $n = 31$:

Table 7: SAT phase transition metrics for rot2 symmetry.

n	Solutions	$\binom{2n}{3}$	$\binom{2n}{3}/((n^2-1)/2)$
23	3,967	15,180	57.5
25	8,980	19,600	62.8
27	17,332	24,804	68.1
29	44,828	30,856	73.5
31	**0**	37,820	**78.8**
33	0	45,760	84.1

The threshold lies at a constraint density of approximately 74 triples per available pair. This is a classic SAT phase transition analogous to the well-known random k-SAT threshold.

We have proven that the collinearity constraint for rot2 on odd n admits a clean algebraic characterization: two antipodal pairs $\{(i_1, j_1), (n-1-i_1, n-1-j_1)\}$ and $\{(i_2, j_2), (n-1-i_2, n-1-j_2)\}$ create a collinear triple *iff*

$$(i_1 - m)(j_2 - m) = (i_2 - m)(j_1 - m),$$

where $m = (n-1)/2$ and the grid coordinates are $0, \ldots, n-1$. Geometrically, this means the two pairs share the same reduced direction from the grid center. The rot2 UNSAT at $n = 31$ thus reduces to a direction-uniqueness conflict: at this threshold, the degree constraints (exactly 2 points per row/column) force a set of n pairs whose directions cannot all be distinct. This equivalence is itself a non-trivial result; proving that it becomes unsatisfiable at $n = 31$ is a SAT phase transition problem comparable in difficulty to proving the random k-SAT threshold.

For $n \geq 33$, only rct4 solutions remain, and **rct4 solutions invariably have the grid center as a circumcenter**. This is a group-theoretic necessity: D_4 orbits on odd-n grids force ≥ 4 points per distance ring.

Thus, missing-center solutions are extinct for all odd $n \geq 33$. The center is always a circumcenter in every maximal $2n$-point solution above this threshold.

Remark on the rct4 gap. An interesting anomaly is that $n = 11$, $n = 13$, and $n = 15$ have *no* known rct4 solutions in the Flammenkamp database, while smaller ($n = 9$) and larger ($n = 17, 19$) values do. This is not a theoretical impossibility but a search gap: the distance ring capacities for $m = 5, 6, 7$ ($n = 2m + 1$) fall into a critical range where rct4's concentrated ring usage pattern cannot be satisfied. The gap could potentially be filled by a dedicated rct4-targeted search.

9.2.2 D_4 Full Reconstruction and Quantitative Model

The Flammenkamp database stores D_4-inequivalent solutions, each representing an orbit under the full dihedral group of the square. We verified the D_4 orbit multipliers empirically (by applying all 8 group transformations to sampled solutions) and reconstructed full solution counts. The multipliers are: 8× for identity class, 4× for rot2 and dia1, 2× for rot4, and 1× for rct4. Cross-validation against our exhaustive C++ enumeration ($n \leq 13$) confirms the reconstruction is exact (all match within sampling error $< 0.1\%$).

The reconstructed full counts enabled a quantitative three-factor weighted least squares model:

$$\text{missing rate} = 16.09 + 0.19 \cdot (\text{mod4}) + 4.08 \cdot (\text{prime}) - 2.29 \cdot \log(\text{ring pairs}),$$

with $R^2 = 0.620$. Surprisingly, the mod4 direct effect is negligible (+0.19), while primality contributes a significant +4.08 — prime n have systematically higher missing rates when controlling for constraint density. The negative log(ring pairs) coefficient reflects the suppression of missing-center solutions in denser collinearity graphs.

Table 8: Reconstructed full missing-center counts.

n	Full Total	Full Missing	Rate	Type
7	132	4	3.03%	Odd
8	380	0	0.00%	Even
9	365	8	2.19%	Odd
10	1135	0	0.00%	Even
11	1120	36	3.21%	Odd
12	4348	52	1.20%	Even
13	3622	292	8.06%	Odd
14	10568	84	0.79%	Even
15	30634	2716	8.87%	Odd
16	46304	1392	3.01%	Even
17	55573	3872	6.97%	Odd
18	152210	24	0.02%	Even
19	258170	10280	3.98%	Odd
20	941580	112	0.01%	Even
21	9701	768	7.92%	Odd
22	5082	52	1.02%	Even

9.2.3 Refined Model: Sum-of-Two-Squares Theory

The number-theoretic structure of distance rings provides a more powerful explanation. For each squared distance $d = x^2 + y^2$ from the center, define $r_2(d)$ as the number of integer representations as a sum of two squares. By Fermat's theorem:

$$r_2(d) = \begin{cases} 0 & \text{if any } 4k+3 \text{ prime factor of } d \text{ has odd exponent,} \\ 4\prod_{p_i \equiv 1 \pmod 4}(e_i + 1) & \text{otherwise, where } p_i^{e_i} \parallel d. \end{cases}$$

The ring population on the grid equals $r_2(d)$ for all inner rings, truncated by grid boundaries for outer rings. For the missing-center problem (requiring ≤ 2 points per ring), rings with $r_2(d) \leq 2$ are the only usable ones.

A refined weighted least squares model incorporating r_2-based features achieves $R^2 = 0.880$, dramatically improving on the raw model:

$$\text{missing rate} = 1.54 - 0.14 \cdot (\text{rings}) - 0.01 \cdot (\mathrm{r}_2 = 0) + 5.01 \cdot (\max 4k + 1 \text{ primes})$$

The dominant factor is the maximum number of $4k+1$ prime factors across all distance rings (coefficient +5.01). This makes sense: $4k+1$ primes generate representable distance values, creating more ring diversity and more opportunities to satisfy the ≤ 2 constraint while maintaining $2n$ total points. The dramatic R^2 improvement (from 0.620 to 0.880) confirms that number theory – specifically the sum-of-two-squares structure of distance rings – is the primary driver of missing-center abundance.

Our analysis suggests that $n = 71$ must be targeted via the **rct4** symmetry class. All known solutions for odd $n \geq 33$, including Heule's $n = 65, 67, 69$, are rct4. The rct4 class reduces the search to selecting fundamental-domain orbits (a polynomial reduction in variable count) and is the only viable structural class for large odd-n grids.

Conjecture 9.1. $D(71) = 2 \times 71 = 142$, *and an rct4-symmetric solution exists. Any such solution necessarily has the grid center as a circumcenter.*

10 Conclusion

We have introduced the missing-center invariant for the No-Three-In-Line problem and provided:

1. A highly optimized forbid-accumulator algorithm ($O(k^2) \to O(1)$ per placement).
2. A rigorous proof of the C_4 theorem (rot4 symmetry $\implies$ center is circumcenter).
3. Complete missing-center counts for $n = 2 \ldots 13$ (full search) and $n = 7 \ldots 45$ (D_4-inequivalent), revealing a three-phase evolution: abundance ($n = 7$–19), decline ($n = 21$–29), and extinction ($n \geq 33$).
4. The rot2 structural phase transition at $n = 31$ (44,890 $\to$ 72, a 600$\times$ collapse).
5. A three-factor model explaining missing-center abundance via parity, primality, and collinearity density.
6. Evidence that $D(n) = 2n$ for all even n and a revised search strategy for $n = 71$ targeting rct4 symmetry.

The code and data are publicly available at `https://github.com/duchenyu/no3inline-missing-center` (version v1.0).

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